WORK SHEET - 14 **FLUID DYNAMICS**

[SINGLE CORRECT CHOICE TYPE]

Q.1 A wire of mass 1 gm is kept horizontally on the surface of water. The minimum length of wire that does not break the surface film is (given surface tension of water is 70 dyne/cm) ($g = 9.8\ m/s^2$)
(A) 3 cm (B) 4 cm (C) 7 cm (D) 14 cm

Q.2 A spherical soap bubble of radius 1 cm is formed inside another of radius 3 cm. The radius of a single soap bubble which maintains the same pressure difference as inside the smaller and outside the larger soap bubble is
(A) 0.75 cm (B) 0.75 m (C) 7.5 cm (D) 7.5 m

Q.3 A straight piece of wire 5 cm long is placed horizontally on the surface of water and is gently pulled up. It is found that a force of 728 dyne in addition to the weight of the wire is required for the purpose. What is the surface tension of the water ?
(A) 0.72 dyne/cm (B) 72.8 dyne/cm (C) 72.8 N/m (D) 72.8 dyne/m

Q.4 There is a horizontal film of soap solution. On it, a thread is placed in the form of a loop. The film is pierced inside the loop and thread becomes a circular loop of radius R. If the surface tension of the solution be T, then what will be the tension in the thread ?
(A) $\pi R^2/T$ (B) RT (C) $2\pi RT$ (D) 2RT

Q.5 The radius of a soap bubble is r. The surface tension of soap solution is T. Keeping temperature constant, the radius of the soap bubble is doubled, the energy necessary for this will be
(A) $24\pi r^2T$ (B) $8\pi r^2T$ (C) $12\pi r^2T$ (D) $16\pi r^2T$

Q.6 There is a small hole in a hollow sphere. The water enters in it when it is taken to a depth of 40 cm under water. The surface tension of water is 0.07 N/m. The diameter of hole is
(A) 7 mm (B) 0.07 mm (C) 0.0007 mm (D) 0.7 m

Q.7 A U-tube with limbs of diameters 5 mm and 2 mm contains water of surface tension 7×10^{-2} newton per meter, angle of contact zero and density 10^3 kg per meter^3. If g is 10 m/sec^2, then the difference in levels in the two limbs is
(A) 8.4 mm (B) 8.4 cm (C) 8.4 m (D) 0.84 mm

Q.8 A steel wire is suspended vertically from a rigid support when loaded with a weight in air, it extends by l_a and when the weight is immersed completely in water, the extension is reduced to l_w. Then the relative density of the material of the weight is

(A) $\frac{l_a}{l_w}$ (B) $\frac{l_a}{l_a - l_w}$ (C) $\frac{l_w}{l_a - l_w}$ (D) $\frac{l_w}{l_a}$

Q.9 Water flows steadily through a horizontal pipe of variable cross-section. If the pressure of water is p at a point where the velocity of flow is v, at another point (pressure p'), where the velocity of flow in ηv: The following statements are given below

(a) $\frac{p'}{p} < 1$, where $\eta < 1$ (b) $\frac{p'}{p} < 1$, where $\eta > 1$ (c) $\frac{p'}{p} > 1$, where $\eta < 1$ (d) $\frac{p'}{p} > 1$, where $\eta > 1$

Choose the correct statement.
(A) (a) and (d) are true.
(B) (a) and (b) are true.
(C) (b) and (c) are true
(D) (c) and (d) are true.

Q.10 A cylindrical vessel is filled with water upto height h. The water flows out in time t sec. If the water is now filled upto heigh ηh, water flow out in time

(A) ηt sec (B) $\eta^2 t$ sec (C) $\sqrt{\eta}$ t sec (D) t/η sec

Q.11 A hollow sphere and a solid sphere of the same outer radius and made of same material are dropped from rest through air from same height. Which will have more velocity just before striking the ground? Neglect viscous force due to air.

(A) Solid sphere (B) Spherical shell
(C) Same for solid and spherical shell (D) Data insufficient

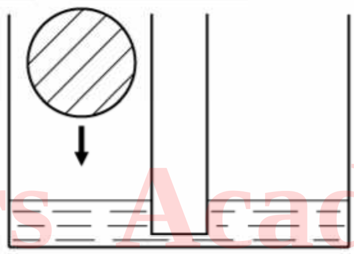
Q.12 A horizontal right angle pipe bend has crosssectional area = 10 cm^2 and water flows through it at speed = 20 m/s. The force on the pipe bend due to the turning of water is :

(A) 565.7 N (B) 400 N (C) 20 N (D) 282.8 N

[PARAGRAPH TYPE] (Multiple correct choice type)

Paragraph for question nos. 13 to 15

In two identical communicating vessel we poured water (see picture). In one of them we put an ice ball of volume V = 100 cm^3 which gets exactly half immersed in the water . The density of water ρ_w = 1000 kg/m^3, the density of ice ρ_i = 900 kg/m^3.



Q.13 Select the correct statement(s). Soon after placing the ice ball in left vessel.

(A) The volume of water flowing to the right vessel is 25cc.
(B) The volume of water flowing to the right vessel is 50cc.
(C) The ice ball is resting on the bottom of vessel.
(D) The ice ball is floating on the water surface.

Q.14 After a long time, select the correct statement(s)

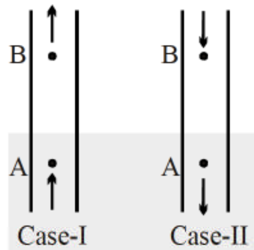
(A) If ice melts, the level of water in both vessel will rise.
(B) If water freezes, such that ball's radius increases the level of water in both vessel will fall.
(C) If ice melts, the level of water in left vessel will rise and level of water in right vessel will fall.
(D) If water freeze such that ball radius increases the level of water in left vessel will fall and level of water in right vessel will rise.

Q.15 Which of the following actions will raise the water level.

(A) Heating the system
(B) Putting a coin on top of ice ball.
(C) Putting a coin in right vessel
(D) Accelerating the system upwards.

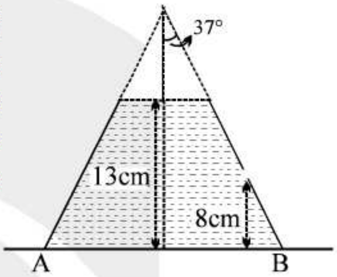
[MULTIPLE CORRECT CHOICE TYPE]

Q.16 Two tubes of uniform cross-section are held vertically. v_A and v_B are the velocities of fluid flow at A and B respectively, and p_A and p_B are pressure at A and B respectively. Arrow show the direction of fluid flow.



(A) In case I, $v_A > v_B$ & $p_A > p_B$ (B) In case I, $v_A = v_B$ & $p_A > p_B$
(C) In case II, $v_A > v_B$ & $p_A < p_B$ (D) In case II, $v_A = v_B$ & $p_A > p_B$

Q.17 Curved surface of a vessel has shape of a truncated cone having semi vertex angle 37°. Vessel is full of water (density $\rho = 1000$ kg/m^3) upto a height of 13 cm and is placed on a smooth horizontal plane. Upper surface is opened to atmosphere. A hole of 1.5 cm^2 is made on curved wall at a height of 8 cm from bottom as shown in figure. Area of water surface in the vessel is large as compared to the area of hole.



(A) Initial velocity of efflux is 1 m/sec
(B) Initial horizontal range of water jet from point B is 6.65 cm
(C) Horizontal force required to keep the vessel in static equilibrium is 0.15 N
(D) Horizontal force required to keep the vessel in static equilibrium is 0.12 N.

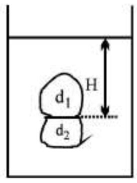
Q.18 A long capillary tube is inserted in water vertically. Water rises to a certain length of the tube. If we want to increase the length for which water will rise, we can
(A) mix some soap solution in water.
(B) raise the temperature of water.
(C) place the whole apparatus in an elevator accelerating downwards.
(D) tilt the tube by a small angle.

Q.19 The diagram below shows a cross-sectional view of a cylindrical pipe of varying diameter.

If an ideal fluid is flowing through the pipe, find the true statement(s).
(A) Volume flow rate is same at A & B
(B) The pressure is lower at point B than at point A.
(C) The volume flow rate is greater at point A than at point B.
(D) The flow speed is greater at point B than at point A.

[INTEGER TYPE]

Q.20 Two objects of equal volume V=1m^3 and different densities $d_1 = 500$ kg/m^3 and $d_2 = 1000$ kg/m^3 are glued to each other so that their contact surface is flat and has an area A = 0.1 m^2. When the objects are submerged in a certain liquid, they float in stable equilibrium, the contact surface being parallel to the surface of the liquid (see the diagram). How deep (H in meters) can the contact surface be in the liquid so that the objects are not torn apart? The maximum force that the glue can withstand is F=250 N.(Neglect atmospheric pressure) **[5]**

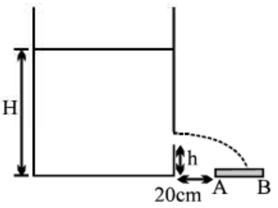


Q.21 An open cylindrical vessel of large cross-section A contains liquid upto a height H = 120cm. After an orifice of area A/1000 at a height of h =20cm is opened.

(a) Calculate liquid heights above orifice for which it falls on both ends of horizontal plate.

(b) How long will the liquid be falling on the plate.

Given : plate AB is of length 60cm. ($g = 10 m/s^2$)



Q.22 The lower end of a long capillary tube of diameter 2 mm is dipped so that its lower end is 10 cm below the surface of water in a beaker.

(a) What is the pressure required in the tube in order to blow a hemispherical bubble at its end in water?

(b) Calculate the capillary rise in the tube if its upper end is left open to atmosphere.

The surface tension of water at temperature of the experiment is 7.30×10^{-2} N m^{-1}, 1 atmospheric pressure = 10^5 Pa, density of water = 1000 kg m^{-3}, angle of contact = 0°

ANSWER KEY

Q.1	C	Q.2	A	Q.3	B	Q.4	D	Q.5	A	Q.6	B	Q.7	A
Q.8	B	Q.9	C	Q.10	C	Q.11	A	Q.12	A	Q.13	AC	Q.14	AB
Q.15	AC	Q.16	B	Q.17	AD	Q.18	CD	Q.19	ABD				

Q.20 H=3m Q.21 (a) 80cm, 5cm; (b) 300sec. Q.22 (a) 1.01146×10^5 (b) 14.6 mm